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# EXERCISE 5: PROCESSING OF AMPLITUDE MODULATION

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## 1. INTRODUCTION

Time is a very important dimension in hearing as almost all real sounds fluctuate over time. The information in sounds is often carried in the *changes* over time rather than in the steady state parts. Temporal processing in the auditory system is discussed in terms of temporal resolution (acuity) and temporal integration (summation). While temporal integration refers to the ability to sum up information over time in order to discriminate stimuli, temporal resolution refers to the ability to detect changes in sounds, e.g., a brief gap in broad-band noise or the presence of amplitude modulation in a noise signal. This exercise focuses on temporal resolution. As seen in Exercise 4 (Frequency Selectivity and Masking), many aspects of human detection performance relating to frequency selectivity and masking can be explained in terms of very early stages of auditory processing, i.e., the properties of the basilar membrane. When discussing temporal resolution, however, central processes in the brain, i.e.stages higher than the auditory nerve, need to be taken into account.

### 1.1. Exercise overview

In this exercise, the temporal resolution of the human auditory system will be measured in terms of modulation detection performance. In the first part, you will carry out a modulation detection experiment with broadband noise carriers and run simulations with the low-pass filter by [1]. In the second part, you will analyse the physical properties of the stimuli. Here it is important to distinguish between the rapid amplitude changes of sounds (*fine structure*) and the slower overall amplitude changes (*envelope*), as illustrated in Figure 1. In the last part, you will conduct another modulation detection experiment, this time with narrowband noise carriers. You will try to predict your measurement results with two different modelling approaches, the low-pass filter model [1] and the modulation filterbank model [2], and analyse the effects of the carrier signal on the detectability of the amplitude modulation.

### 1.2. Exercise objectives

After completion of this lab exercise and the associated report it is expected that you should be able to:

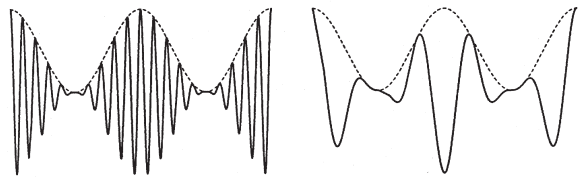

**Fig. 1**. Illustration of fine structure vs. envelope of a sound. Both panels show the result of applying a sinusoidal amplitude modulation to a pure-tone carrier with a modulation depth $m = 1$ (fully modulated). On the left, $f_c = 10 \cdot f_m$ and on the right $f_c = 2.5 \cdot f_m$, where $f_c$ denotes the carrier frequency and $f_m$ the modulation frequency. The dashed line is the envelope of the sound, the solid line is the fine structure. From [3].

- Describe the characteristics of a sinusoidal amplitude modulated signal in the time- and frequency domain.
- Explain the effect of the carrier signal on the detectability of the amplitude modulation.
- Relate the concepts of signal waveform, spectrum, envelope and envelope spectrum.
- Describe the properties of the temporal modulation transfer function.
- Explain the concept of a low-pass model of auditory temporal processing and its deficits.
- Explain the concept of a modulation filterbank model of auditory temporal processing as an alternative to the low-pass model.

### 1.3. Exercise background

One method of investigating temporal processing is to measure the detectability of sinusoidal amplitude modulation as a function of the modulation frequency. A sinusoidally amplitude-modulated signal is described by:

$$s(t) = n(t)(1 + m\cos(2\pi f_m t)) \,, \tag{1}$$

where $s(t)$ is the modulated signal, $n(t)$ is the carrier waveform, $m$ is the modulation depth, and $f_m$ is the modulation frequency. Often bandpass-filtered Gaussian noise is used

as the carrier, but in principle any carrier can be used. The detection threshold is measured by varying the modulation depth $m$ of the imposed amplitude modulation. The threshold is reached when the modulation is just detectable. The temporal modulation transfer function (TMTF) is defined as the modulation depth at threshold, usually expressed in dB (i.e., $20log_{10}m$), as a function of the modulation frequency.

### 1.4. Exercise setup

For this exercise, the following statements should be entered after starting Matlab and going to the `PoAM` folder in order to be able to access the provided scripts:

```
startup
% sets the path
```

Follow the instructions below to set-up the audio playback:

1. Connect a pair of headphones or earphones to your computer, preferably using a cable.
2. Ensure that the sound output of your computer is configured to your headphones and that the volume is adjusted to a comfortable level.

   **IMPORTANT**: Always check the volume levels on your computer ***before*** putting the headphones on in order to ensure that the output level is at a safe level for your ears!

## 2. PART 1: MODULATION DETECTION WITH BROADBAND NOISE CARRIERS

[1] measured TMTFs for a broadband noise carrier (6 kHz bandwidth). The results are shown in Figure 2.

In this part of the exercise you will reproduce this experiment using a broadband noise carrier signal, plot the results, and relate them to the data depicted in Figure 2. Further, you will simulate the same experiment using a simple low-pass model as a "test subject" in order to find the best match for your data and interpret the model parameter (the cutoff frequency of the low-pass filter).

### 2.1. Experiment

A TMTF experiment, called `modulationDetection1`, is provided within the MATLAB AFC framework. See Appendix A for a description of how to start an experiment and how to plot and access the data. The experiment uses a 1-up 2-down three-alternative forced-choice paradigm and is set up with a band-limited noise carrier with a fixed upper cutoff frequency of 4 kHz. The bandwidth of the noise carrier can be chosen. Choose a **bandwidth of 3 kHz** for this experiment, such that the noise is band-limited between 1 kHz and 4 kHz. The presentation level is approximately 70 dB SPL. Each point is measured twice for each subject. Use the AFC script to measure the six points of the TMTF provided by the setup (modulation frequencies of 8, 16, 32, 64, 128, 256 Hz) for two group members. (Experiment duration: approximately **30 minutes per listener**).

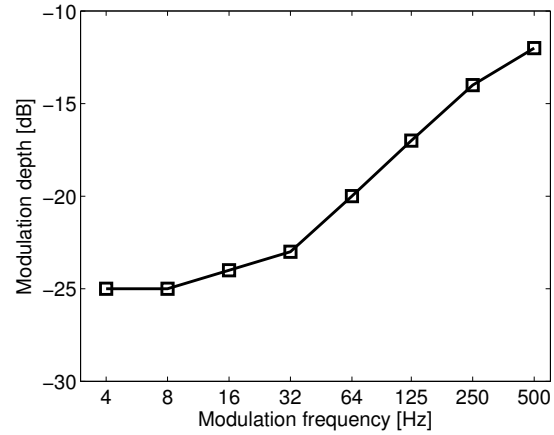


**Fig. 2**. TMTF measured with a broadband noise carrier from [1].

### 2.2. Simulation using low-pass model

Viemeister suggested that the *"sluggishnes"* of the auditory system could account for the data. He developed an amplitude processing model inspired by this assumption, which consists of the four stages depicted in Figure 3. A broadband "predetection filter" restricts the spectrum of the signal to a specific pass band, followed by half-wave rectification representing the inner hair cell transduction in the inner ear and a first-order low-pass filter with a cut off frequency of 64 Hz (time constant $\tau = 2.5$ ms), representing the sluggishness of the auditory system. Effectively, the latter two stages extract the envelope of the signal. The fourth stage calculates the modulation depth of the filter output in a certain time interval. Effects of basilar membrane filtering are ignored as they do not play a crucial role for amplitude modulation detection.

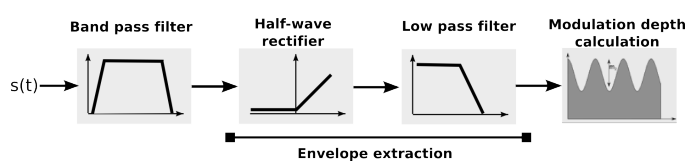


**Fig. 3**. Schematic of Viemeister low-pass modeling approach

The model `lowPassModel` is an implementation of Viemeister's low-pass model which runs in the AFC framework. See Appendix A for information about running the experiment with a model instead of a person. The decision

device selects the interval with the largest modulation depth from the three alternatives.

The model is non-deterministic in the sense that its output depends on the noise. For this reason, all modulation detection simulations are automatically run three times in this exercise so that thresholds can be estimated more precisely. If the model cannot detect full modulation ($m = 1$) for a given condition it outputs the text "track skipped". In this case, the AFC program gives a "dummy value" of 0.1234 dB as the resulting threshold. The file `lowPassModel_init` in the directory `scripts\models` contains the cutoff frequency of the first order low-pass filter used in the model.

### 2.3. Questions

- **[RESULTS]** Plot the measurement results using the supplied MATLAB script, `pldata`. Note that in the plot modulation depth $m$ is expressed in dB (i.e., $20 \log_{10} m$). The modulation frequency axis should be scaled logarithmically (e.g., using `set(gca,'XScale', 'log'))`).
- **[RESULTS]** Describe the shape of the measured TMTF and compare it to the data in Figure 2.
- **[IMPLEMENT]** Run the low-pass model and try to find a cutoff frequency value such that the resulting TMTF resembles the TMTF you measured yourself. You may have to run the model several times. Which cutoff frequency matches your data the best?

  HINT: In order to rerun the model you need to delete or rename the two files `control_modulationDetection1 lowPassModel_3000_groupX`, which contains the control information, and `modulationDetection1 lowPassModel_3000_groupX`, which contains the results given by the model.
- **[RESULTS]** Plot the TMTF predicted by the model when you are satisfied with the match.
- **[RESULTS]** What is the time constant corresponding to the filter that gives the best match? This value can be considered as a measure of the temporal resolution of the auditory system.

## 3. PART 2: EXAMINING THE PHYSICAL PROPERTIES OF THE STIMULUS

Earlier in the course, the physical properties of stimuli were examined in the time and frequency domain. When investigating temporal resolution, the *envelope* of the stimuli must be examined in addition to this. The [1] model extracts the envelope using half-wave rectification and low-pass filtering. Alternatively, and mathematically well-defined, the envelope can be extracted by use of the Hilbert transform.

Given a real signal $s(t)$, the corresponding *analytic signal* is given by $s_a(t) = s(t) - \mathrm{i}\, S_H(t)$, where $S_H(t)$ denotes the Hilbert transform of $s(t)$:

$$S_H(t) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{s(\tau)}{\tau - t}\, d\tau \,. \tag{2}$$

The Hilbert-envelope $h_{\text{env}}(t)$ is then given as the modulus of the analytic signal:

$$h_{\text{env}}(t) = |s_a(t)| = \sqrt{s^2(t) + S_H^2(t)} \,. \tag{3}$$

The envelope spectrum or "modulation spectrum" is defined as the power spectrum of the Hilbert envelope.

As introduced earlier, a sinusoidally amplitude-modulated signal is described by

$$s(t) = n(t)(1 + m \cos(2\pi f_m t)) \,, \tag{4}$$

where $s(t)$ is the modulated signal, $n(t)$ is the carrier waveform, $m$ is the modulation depth, and $f_m$ is the modulation frequency. Often bandpass-filtered Gaussian noise is used as the carrier, but in principle any carrier can be used.

### 3.1. Questions

- **[IMPLEMENT]** Write a MATLAB script that generates bandpass filtered noise with bandwidths of 3 Hz, 30 Hz, 300 Hz and 3 kHz, all with an upper cutoff frequency of 4 kHz and duration of at least 1 second.

  HINT: The MATLAB function `bpnoise` is supplied. Type `help bpnoise` to see how to use it.
- **[RESULTS]** Calculate and plot the temporal waveform, the power spectrum, the envelope, and the envelope power spectrum for each of the noise files.

  HINTS: MATLAB has a built-in function `x = hilbert(xr)`, that calculates the *analytic signal* from a real-valued data sequence. Note that due to the random nature of the noise signal, only the *average* envelope power spectrum across a large number noise signals converges to the true power spectrum. Importantly, this averaging must be done **after** taking the absolute value of the FFT.
- **[DISCUSS]** Describe and explain, in your own words, the link between the plots (temporal wave form, the power spectrum, envelope and envelope power spectrum).
- **[IMPLEMENT]** Write a MATLAB script that generates similar bandpass filtered noise as described above, but this time with an imposed sinusoidal modulation of 16 Hz.

- **[RESULTS]** Plot the power spectra and envelope power spectra of the modulated noise. Compare them to the previous power spectra and envelope power spectra without imposed sinusoidal modulation.

- **[DISCUSS]** What effect does applying sinusoidal amplitude modulation have on the spectrum of a signal?

- **[DISCUSS]** What effect does applying sinusoidal amplitude modulation have on the overall power? HINT: The power of a continuous signal is defined as:

$$P_s = \lim_{T\to\infty} \frac{1}{T} \int_0^T s^2(t)\,\mathrm{d}t. \tag{5}$$

- **[DISCUSS]** What detection cues besides temporal changes might be available to a test subject?

- **[DISCUSS]** Why is broadband noise often preferred as a carrier over narrow-band noise or pure-tones?

## 4. PART 3: MODULATION DETECTION WITH NARROW-BAND NOISE CARRIERS

[2] measured TMTFs for carrier bandwidths of 3 Hz, 31 Hz and 314 Hz. The results are shown in Figure 4.

In this part of the exercise, you will measure your modulation detection performance for the 10 Hz point in Figure 4, i.e., for only one modulation frequency but three different noise carrier bandwidths. Further on, you will simulate the experiment for a range of modulation frequencies using (i) the Viemeister low-pass model and (ii) the Dau filterbank model as "test subjects" and compare the results to the data in Figure 4.

### 4.1. Measurement of modulation detection performance

A TMTF experiment, called `modulationDetection2`, is provided within the MATLAB AFC framework. The only difference to `modulationDetection1`, used in section 2, is the range of modulation frequencies used. See Appendix A for a description of how to start an experiment and how to plot and access the data. The experiment uses a 1-up 2-down three-alternative forced-choice paradigm and is set up with a band-limited noise carrier with a fixed upper cutoff frequency of 4 kHz. The bandwidth of the noise carrier can be chosen. The presentation level is approximately 70 dB SPL. Each point is measured twice for each subject. Use the AFC script to measure one point of the TMTF provided by the setup (modulation frequency of 10 Hz) for two group members. Run the experiment for bandwidths of 3 Hz, 30 Hz, and 300 Hz. (Experiment duration: approximately **15 minutes per listener**).

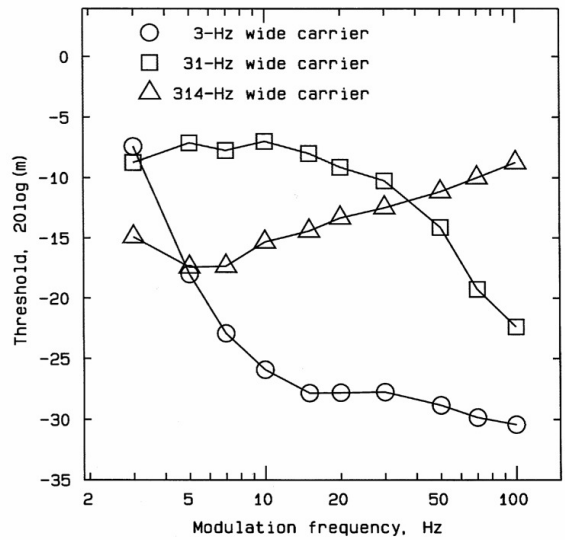


**Fig. 4**. The TMTF for carriers with different bandwidths as measured by [2]. This replot is from [4, Chap. 5].

### 4.2. Simulations using the low-pass model and the filterbank model

#### *4.2.1. The low-pass model*

An adapted version of the Viemeister low-pass model used in section 2 is provided (`lowPassModel2`). The file `lowPassModel2_init` in the directory `scripts\models` contains the cutoff frequency of the first order low-pass filter used in the model. Use the cutoff frequency you determined in section 2.

#### *4.2.2. The filterbank model*

Based on the data depicted in Figure 4, a modulation filterbank modeling approach for modulation detection was suggested in [2] and [5]. A schematic of the modeling approach described in [5] is depicted in Figure 5. The first stage is an envelope extraction using the Hilbert transform, the second stage consists of a bandpass filter which is *tuned to the imposed modulation frequency*, and the third stage calculates the modulation power at the output of the filter. The filters have a fixed bandwidth of 5 Hz for filter center frequencies smaller than 10 Hz and a constant Q of 2 for frequencies equal to or greater than 10 Hz.

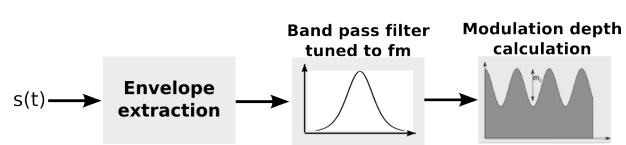


**Fig. 5**. Schematic of Dau filterbank modeling approach

The model `ModFilterBankModel` is an implementation of the Dau filterbank model described above, which runs in the AFC framework. As for the low-pass model, the decision device selects the interval with the largest modulation depth from the three alternatives.

### 4.3. Questions

- **[RESULTS]** Extract the measurement results using the supplied MATLAB script, `pldata`.

  HINT: Note that you measured only one data point per bandwidth and subject. The results for each bandwidth are stored separately. The plot automatically generated by `pldata` is therefore not an appropriate depiction. Instead, directly extract the measured values for the individual bandwidths using `pldata` as described in Appendix A and make a plot that combines the results of all listeners and for all bandwidths.

- **[RESULTS]** Compare your results to the data depicted in Figure 4. For plotting, the data from [2] are available as a `.mat` file in the data folder.

- **[DISCUSS]** Try to explain the perceptual differences for the individual noise carrier bandwidths based on your impression during the experiment.

- **[IMPLEMENT]** Run experiment `modulationDetection2` with the model `lowPassModel2` for bandwidths of 3 Hz, 30 Hz, 300 Hz, and 3 kHz. The experiment will automatically run for 10 points (modulation frequencies of 3, 5, 7, 10, 15, 20, 30, 50, 70, and 100 Hz), using 3 repetitions for each point.

- **[RESULTS]** Plot the results and compare them to the data depicted in Figure 4 and Figure 2, respectively.

- **[IMPLEMENT]** Run the `modulationDetection2` experiment with the `ModFilterBankModel` model for bandwidths of 3 Hz, 30 Hz, 300 Hz, and 3 kHz. The experiment will automatically run for 10 points (modulation frequencies of 3, 5, 7, 10, 15, 20, 30, 50, 70, and 100 Hz), using 3 repetitions for each point.

- **[DISCUSS]** Do you know a comparable concept to the modulation filterbank model in another domain of auditory signal processing?

- **[RESULTS]** Use `pldata` to extract the modeled data and plot the data according to Figure 4, i.e., plot the four TMTFs for the individual bandwidths on top of each other using appropriate symbols and a legend. The modulation frequency axis should be scaled logarithmically (e.g., using `set(gca,'XScale', 'log')`).

- **[RESULTS]** Compare the modeled data to the data depicted in Figure 4 and Figure 2, respectively. The data from [2] are also available as a `.mat` file in the data folder and can be used for comparison with the modeled results.

- **[DISCUSS]** Provide a detailed discussion of the results obtained using the low-pass model and the filterbank model based on the envelope spectra you plotted in section 3.

## 5. REFERENCES


[1] N.F. Viemeister, "Temporal modulation transfer functions based upon modulation thresholds," *J. Acoust. Soc. Am.*, vol. 65, pp. 1364–1380, 1979.

[2] T. Dau, B. Kollmeier, and A. Kohlrausch, "Modeling auditory processing of amplitude modulation. i. modulation detection and masking with narrowband carriers," *J. Acoust. Soc. Am.*, vol. 102, pp. 2892–2905, 1997.

[3] N.F. Viemeister and C.J. Plack, *Human Psychophysics*, Springer-Verlag, New York, 1993.

[4] Brian C. J. Moore, *An Introduction to the Psychology of Hearing*, Academic Press, 5th edition, 2003.

[5] Torsten Dau, Jesko L. Verhey, and A Kohlrausch, "Intrinsic envelope fluctuations and modulation-detection thresholds for narrow-band noise carriers," *J. Acoust. Soc. Am.*, vol. 106, no. 5, pp. 2752–60, 1999.

# A. AFC MATLAB TOOLS

## A.1. Starting measurements

To start a measurement, use the `afc_main` command:

```
afc_main(expname, ts, userpar, group);
```

The input parameters are as follows:

- `expname:` The name of the experiment
- `ts:` The initials of the test subject
- `userpar:` User parameter. Here the bandwidth of the noise carrier (e.g. 3000 Hz)
- `group:` Your group number

For example, to start the experiment `modulationDetection1` for subject 'John Doe', member of group 3, with a carrier bandwidth of 3 kHz enter the following command:

```
afc_main('modulationDetection1','jd','3000','group3');
           % This is just an example!
```

You can use the numeric keypad to answer the questions during the measurement, just make sure that 'NumLock' is activated (check the NumLock-LED).

It is possible to substitute the subject name with the name of a model. In this case, the model will provide the response instead of a human test subject. The rest of the procedure is the same as when a human test subject is used. For example, to start the experiment `modulationDetection1` for the model `lowPassModel`, with a 3 kHz-wide bandlimited carrier, as a member of group 3, enter the following command:

```
afc_main('modulationDetection1','lowPassModel','3000','group3');
           % This is just an example!
```

The answers from the model are based on MATLAB scripts which are placed in the `model` folder .

## A.2. Plotting the results

The data of all the measurements will be saved in your current working directory.[1]. The different filenames follow the scheme

```
expname_vp_userpar_group.dat
```

Thus, the measured data for the example given above could be found in the file `modulationDetection1_jd_3000_group3.dat`. To plot the results of AFC measurements, use the command `pldata`:

```
[x,y,ystd] = pldata(expname, vp, userpar, group, proc);
```

The input parameters are almost the same as described above:

- `expname:` The name of the experiment
- `vp:` The initials of the subject who did the measurement. To see results of more than one subject, enclose the initials with curly brackets. `{}`. Examples:
  - `'jd'`
  - `{'jd', 'bm', 'co'}`
- `userpar:` The value of the condition used. Example:
  - `'3000'`
- `group:` The name of your group
- `proc:` Optional Parameter. Possible values:

[1] If you changed the directory (e.g., by using the `cd` command) you can get the actual working directory with the `pwd` command.

- `'mean'` Plot the mean data
- `'raw'` Plot the raw data (default)

The (optional) output parameters are:

- `x:` The experimental parameter; in this case, the carrier bandwidth
- `y:` The results of the experiment; in this case, modulation depth at threshold in dB or gap width
- `ystd:` The standard deviation of `y`